

FINAL: GROUP THEORY

Date: **10th November 2025**

The Total points is **115** and the maximum you can score is **100** points.

- (1) (3+3+3+9=18 points) Define a group action. State the Orbit-Stabilizer Theorem. What is a conjugacy class in a group? Show that the set of all 5-cycles in A_5 is the union of two conjugacy classes.
- (2) (3+12=15 points) Define the center of a group and compute the center of $SL_3(\mathbb{F}_7)$ where $\mathbb{F}_7 = \mathbb{Z}/7\mathbb{Z}$ is the field containing 7 elements.
- (3) (10+5=15 points) Let G be a group and $H, K \leq G$ such that $G = HK$, H normal in G , and both H and K solvable. Prove that G is solvable. Is the result true if H is not normal in G ?
- (4) (12 points) Write down explicitly a subgroup of S_8 that is isomorphic to the quaternion group Q_8 . Give a proof.
- (5) (5+10=15 points) Define a simple group. Show that a group of order 462 is not simple.
- (6) (5+15=20 points) Define (external) semidirect product. Let H be a finite group. Show that there exists a group G such that $H \trianglelefteq G$ and given any automorphism σ of H there exist a $g \in G$ such that $\sigma(h) = ghg^{-1}$ for all $h \in H$.
- (7) (6+6+8=20 points) Let $SO_n(\mathbb{R})$ denote the group of all $n \times n$ real orthogonal matrices of determinant 1.
 - (a) Show that every matrix in $SO_2(\mathbb{R})$ is a rotation matrix about the origin, and explicitly describe its form.
 - (b) Find all eigenvalues of elements in $SO_2(\mathbb{R})$.
 - (c) Find all eigenvalues of elements in $SO_3(\mathbb{R})$.